Reproducibility Supplement

Metric random tilings: exact entropy and the diffraction of counted disorder

RS v1.3 (2026-11-12; the repository distributes code, data and figures only: the internal audit register is no longer included, development working documents that are not registered in Table R2 have been removed, and the pointers to them are replaced by availability-from-the-authors notes. The registered scripts, data, figures and provenance records are unchanged.)

RS v1.2 (2026-11-11; released together with SI v3.0c and the open code repository. RS1 prose figure pointers aligned with the SI v3.0 figure renumbering — the v1.1 revision had updated the Table R2 rows but missed the prose, so its Figs. S8/S9/S10 pointers were one step behind; file paths are now stated in the release-repository layout (`02_figures/`, `03_data/`, `04_scripts_L2/`, `05_predecessor_assets/`, `06_engines/`), and engine data files carry ASCII filenames in the release (file contents are unchanged, byte-identical to the development originals). The entropy-split ideal-mixing erratum carried by SI v3.0c changes no row of this document: the registered scripts and data regenerate the corrected figure as released.)

RS1. The computational register

Exact arithmetic and guards. Honeycomb/3.4.6.4 geometries are built in $\mathbb{Q}(\sqrt{2})$, $\mathbb{Q}(\sqrt{3})$; every matrix reduction is verified by three independent rank computations (Fraction elimination, two modular primes, float SVD); count consumers assert Euler-type identities (V = 2F, E = 3F on tori) before counting.

Counting ladder. Engines are organised in three tiers, ordered by cost and reach; every tier is anchored in the tier below it, and Table R1 maps each engine to its reach, usage, and verification.

Tier 1: exact enumeration. DFS over partial edge assignments with integer kernel parametrisation, prefix-blocked parallelism, and brute-force 3^E on the smallest systems. Reach $F \lesssim 20$; role: verification anchors and small exact counts (honeycomb 2×2–5×2).

Tier 2: transfer matrices, traces, elimination (the exact workhorse).[S10] A keyed dynamic program over partial assignments (states merged by key before vectorisation, since path counts are not state counts) builds the single-column strip operator T. Strips are counted by iterating T; the torus count is the exact trace $\Omega(L_x \times L_y) = \mathrm{Tr}(T^{L_x})$, computed from one operator for all lengths (honeycomb: $|S| \le 3^6 = 729$, sub-second per size, series to $L_x = 20$). At scale T is held in CSR sparse form and the trace is evaluated in chunked unit-vector blocks with per-batch/segment checkpoints and an integrality assert on every partial sum (3.4.6.4: $|S| = 2^{2L+1}$; the L = 10 point = 4 002 790 772 868 took 36 320 s, ~215 GB peak). Systems whose constraints are naturally face-based rather than cut-based (4.6.12) are counted in the same exact tier by constraint-space elimination: bucket elimination over elimination

orderings, and a parameterised DFS with exact rational back-substitution where the bucket width explodes. Closed forms replace counting where theorems provide them (square $\Omega = k^{Lx+Ly}$; Penrose strips $\Omega = k^{\#strips}$; honeycomb binary/non-arithmetic classes $\Omega = k^{Lx+Ly+1}$ on coprime tori) and are verified bitwise at every measured size.

Cylinder spectral radii. Wide-strip and face-constraint systems are adjudicated by the cylinder transfer-matrix spectral radius $\rho(W)$ (the exact asymptotic per-column growth rate, with no finite-length transient), computed on explicit sparse operators with ARPACK (leading spectra are typically degenerate: ± pairs, W-th roots-of-unity families, accidental multiplicities; power iteration is inadmissible). The verdict discipline has four gates, each once omitted at cost: (i) machine-bitwise anchors: frontier DP reproduces exact torus counts, and the full block product from empty head to empty tail equals the exact count; (ii) head-side purity: K-sample true-head closures, excluding unreachable-class inflation; (iii) tail connectivity: reverse-BFS trunk restriction with a restricted Perron computation (a Perron value over states that cannot reach the empty tail overestimates); (iv) known-answer calibration: the whole procedure reproduces the exactly known W = 2 rate with zero deviation. Exact closed-form sub-laws (honeycomb binary/non-arithmetic $\Omega = k^{Lx+Ly+1}$ on coprime tori; the 3.12.12 mid-system 2^{L+W+1}) are transient sectors under the roots-of-unity selection rule (S6.3), bitwise true and asymptotically sub-leading; see S8.1 L4′.

Tier 3: calibrated stochastic estimators. Deployed only where exact counts cap out, and only under an anchor ladder. (a) *Knuth-type unbiased tree sampling*[S9] (two-step lookahead): raw estimates miss by 40–62% at Ω level (heavy-tailed weights) but only ≈ 0.01–0.02 at entropy-density level, and values are calibrated against exact anchors (24-task calibration, deviations +0.0001…−0.0049, statistical error ±0.012; RS1). It is deployed on the 4.6.12 ternary $\mathbb{Z}$ series (Table S6). It becomes ineffective on the 4.8.8 Λ lattice at W ≥ 3: the same heavy tails. (b) *Hutchinson stochastic trace*[S5] estimates $\mathrm{Tr}(T^L) = \langle v^{\mathsf{T}} T^L v \rangle$ over Rademacher probes: unbiased by the trace identity, and reliable precisely where constraint locking compresses the transfer matrix so probe statistics stay near-Gaussian (kurtosis ≤ 3.3 measured across 3.4.6.4 L = 2…8). Deployment requires a calibration pass at known-answer sizes; it fails otherwise (honeycomb binary $L_y = 5$: probe variance diverges). Deployed: 4.8.8 Λ W = 5 (20 000 probes × 8 seeds, ± 0.0022%).

Verification protocol (cross-cutting). (i) An anchor ladder ties every stochastic estimator to exact values at overlapping sizes, and the exact tiers to brute force (three-way DFS/DP/brute agreement on all narrow-strip cases). (ii) Independent code paths (DFS enumeration, keyed dynamic program, sparse trace) agree bitwise at every overlapping size (729/1017/22 767 on the ternary class). (iii) Exact symmetries are asserted at run time: transposition $\Omega(L_x \times L_y) = \Omega(L_y \times L_x)$, mirror-pair equality under 2↔3, state-count laws ($|S| = 2^{2L+1}$ for the 3.4.6.4 cylinder), Euler-type identities (V = 2F, E = 3F) before counting. (iv) Expensive extrapolations are guarded by registered out-of-sample predictions (ln Ω(9), ln Ω(10) hit). (v) Fail-closed budgets: any enumeration whose materialised size

would exceed the registered cap (k^{dim} × row width ≤ 2×10^8; per-worker intermediate-state gates) is refused before allocation; long runs are checkpointed and resumable.

Reproducibility. All randomness is explicitly seeded; every CSV cited regenerates its figure deterministically. Main-figure scripts: `fig1_model_classification.py` (Fig. 1), `fig2_decay_mechanisms.py` (Fig. 2; pattern assets `02_figures/assets_decay/`, data from `06_engines/z4_square/`, `06_engines/z4_kagome/`, `06_engines/z3_488/`), `fig3_honeycomb_metric_entropy.py` (Fig. 3; panels c/d = canonical `fig2cd_symmetric_forms.py` functions, 1:1), `fig4_value_domain_halos.py` together with the S9 suite machinery (`vd_suite3.py`/`vd_patch.py`; data `03_data/uniform/`; Fig. 4) + asset generators (`gen_G_pieces.py`, `gen_K3_kagome_portrait.py`, `gen_R4_rth_colorful.py`, `gen_R4_rth_binary.py`, `gen_G_batch2.py`, `gen_G_batch3.py` in `05_predecessor_assets/`; `gen_S9_torus_construction.py`, `gen_metric_patches.py` in `04_scripts_L2/`) + shared style `l1_style.py`; SI figures: `gen_SI_figs.py` (S1–S5), `figSI_area_law.py` (S6) and `figS7_4612_spectral.py` (S7); Figs. S9 and S10 and the S10.1/S10.2/halo-orientation data are computed on the delivered suite by `vd_si_sigs.py` (`03_data/uniform/pdf_*`, `slq_*`, `halo_orient_*`), and Fig. S11 with its data table by `fig_si_entropy_split.py` and the exact enumerator `si_data.py` (`03_data/uniform/si_entropy_split.csv`), with canonical copies in `04_scripts_L2/` (manifest: Table R2); development originals are kept in the project tree; the released snapshot is openly available at https://github.com/shtusynelab/metric-random-tiling. The value-domain ensemble suite's generation scripts (`vd_suite3.py`, `vd_patch.py`, `vd_torus_run.py`, `vd_wrms.py`, vendored `sampler2.py`/`metric_hc.py`) sit in `04_scripts_L2/` (Table R2), together with the revision-round drivers `vd_ladder_ext.py` (ladder extension and patch-size series) and `vd_tension_v2.py` (tension readouts), and the server drivers `vd_torus_run_v2.py` and `vd2_gate.py` (mirror gate). The planar open-boundary disk counting of S6.3 is reproducible from `04_scripts_L2/disk/`: `disk_count.c` (the production engine, C with `__int128` and OpenMP; scatter and gather twin modes), `disk_lib.py` (reference implementation with the DFS, frontier-DP and column-DP paths and the torus column trace) and `blocked_multi.py` (blocked-DFS cross-check), with the exact counts and fit summary in `03_data/uniform/disk_counts_exact.csv`. The constructive certificates of SI S8.3 are covered by the certificate-checker engine of Table R1, with scripts and records mapped in Table R2.

RS2. Counting engines

Table R1 | Counting engines: reach, usage, and verification.

Engine (tier)	Core technical point	Reach / applicable systems	Used for	Anchored by
DFS + brute force (1)	partial-assignment search; integer kernel parametrisation; prefix-blocked parallelism	$F \lesssim 20$, any network	anchors everywhere; small exact counts	three-way DFS/DP/brute agreement (34/34 strip cases)

Engine (tier)	Core technical point	Reach / applicable systems	Used for	Anchored by
Keyed DP → strip TM (2)	states merged by key before vectorisation (path counts ≠ state counts); one operator, all lengths via $\mathrm{Tr}(T^{L_x})$	strips and tori with compressible cut states	honeycomb series ($L_x \le 20$ at $L_y = 2$, 3, 5; $L_y = 7$, 9 by transposition), narrow strips W = 2–4, 4.8.8 $\mathbb{Z}$ and Λ W ≤ 4	DFS anchors digit-for-digit (729/1017/22 767)
Exact sparse trace (2)	CSR operator; chunked unit-vector blocks; checkpoints + integrality asserts	triangle-locked cut states, $\lvert S\rvert = 2^{2L+1}$	3.4.6.4 L = 6–10	L = 2–5 ≡ Fraction/DP values
Cylinder TM spectral radius (2)	explicit sparse cylinder operator; ARPACK (degenerate leading spectra (± pairs, roots-of-unity families, accidental degeneracies) defeat power iteration); trunk/tail-connectivity + known-answer calibration gates	wide-strip and face-constraint systems: 3.12.12 mid (W = 2…8), honeycomb three classes (W = 2…9), 4.6.12 binary (W = 2…4)	entropy signs and asymptotes (S5.3, S6.2, S8.2)	Gate A/B bitwise anchors; Tr-vs-DP identity; W = 2 trunk calibration (zero deviation)
Constraint-space elimination (2)	bucket elimination over orderings; parameterised DFS with exact rational back-substitution	face-based constraint systems (4.6.12: 12-edge faces, 4 equations/face)	4.6.12 L = 1–2 exact; L = 3, 4 by calibrated Knuth	L = 1 brute force; four affine anchors 64/64/518/22 767
Frontier DP, blocked DFS, strip Knuth (2)	face-closure-ordering keyed DP (states merged by key); prefix-parameterised blocked DFS; strip Knuth with X = Πf weights	4.6.12 binary {0,1}: square/rectangular tori and W×L strips	binary exact to L = 3 tori and 4×3; W = 2 strips to L = 14 (DP); calibrated strip Knuth over W = 2…5 (reach L ≈ 10/5/4 at W = 2/3/4)	anchors 112 / 22 912 / 58 240 / 11 261 320 / 15 300 096 dual-engine; Knuth pinned at four strip points (0.03–2%)
Knuth tree sampler (3)	unbiased self-avoiding tree sampling, two-step lookahead; entropy-density calibration	4.6.12 ternary $\mathbb{Z}$ L = 3, 4	4.6.12 ternary series (Table S6)	24-task calibration vs exact anchors (+0.0001…−0.0049, ±0.012); ineffective on Λ at W ≥ 3
Hutchinson trace (3)	$\mathrm{Tr}(T^L) = \langle v^{\mathsf{T}} T^L v\rangle$, Rademacher; exactly unbiased; kurtosis-monitored	locked/compressed operators with near-Gaussian probe tails	4.8.8 Λ W = 5 (± 0.0022%)	mandatory known-answer calibration pass; fail precedent: honeycomb binary $L_y = 5$
Closed forms [T]	theorems replace counting	square $\Omega = k^{L_x+L_y}$; Penrose $\Omega = k^{\#strips}$; honeycomb binary/non-arithmetic $\Omega = k^{L_x+L_y+1}$ (coprime sizes; the exact sub-law branch, S6.3)	S1.2, S6.2–S6.3	bitwise verification at every measured size
Constructive-certificate checkers (1)	finite-patch double-completion certificates (ternary); stripe-formula vs edge-DFS dual strip counting; interval-criterion vs direct-walk simulation; subset injectivity; negative controls	four carrier patches ({1,2,3}) and binary-honeycomb site sets (W = 3…24)	constructive lower bounds of SI S8.3 (Fig. S8)	exact face-equation checks (rational and √3 parts); bitwise dual counts (72/156/358 anchors); 2^{14} subset tests; square-lattice kernel = 0 control

RS3. File map

Table R2 | File map: paper items to data and code. Scripts without a path prefix are canonical copies in `04_scripts_L2/`; counting-engine scripts are the exception and live beside their data in the per-network directories under `06_engines/` (the Table R1 row gives their paths in full); items marked (dev) are development-stage scripts, archived with this repository snapshot. Paths are repository-relative (`02_figures/`, `03_data/`, `04_scripts_L2/`, `05_predecessor_assets/`, `06_engines/`). References to SI tables (S1–S7) and sections (S1–S11) point to the main SI.

Item	Data / code
Fig. 1 (model & classification)	`02_figures/fig1_model_classification.*`; script `fig1_model_classification.py`; assets `02_figures/assets_carriers/`, a/b/c from the shared asset family (lozenge `G1_lozenge_color.png`, honeycomb `G5b_honeycomb_faces.png`, 3.4.6.4 `G4_rth_colorful.png`; generators `gen_G_pieces.py`, `gen_K3_kagome_portrait.py`, `gen_R4_rth_colorful.py`, `gen_R4_rth_binary.py`, `gen_G_batch2.py`, `gen_G_batch3.py` in `05_predecessor_assets/`) + d/e rendered by `gen_metric_patches.py` (black-edge style)
Fig. 1f counts	`06_engines/z3_hexagonal_P0/P0_PM_torus_growth_forms.csv`; `06_engines/z3_hexagonal_P0/P0_PM_torus_trace_extension_Ly3.csv`; `06_engines/z4_3464/RTH_torus_area_law_8pts.csv`; `06_engines/z3_488/TSQ_Lambda_DP_exp.csv`; `06_engines/z3_31212/TSD31212_dp_ternary.csv`
Fig. 2 (decay mechanisms)	`02_figures/fig2_decay_mechanisms.*`; script `fig2_decay_mechanisms.py`; data `06_engines/z4_square/SQ_entropy_density_verification.csv`, `06_engines/z4_kagome/KAG_entropy_density_verification.csv`, `06_engines/z3_488/TSQ_Lambda_DP_exp.csv`; pattern assets `02_figures/assets_decay/` (G2/G3/G4 from the shared asset family + the drawn triangular patch)
Fig. 3 (honeycomb geometric limit)	`02_figures/fig3_honeycomb_metric_entropy.*` (file named to the main-text number); script `fig3_honeycomb_metric_entropy.py` (panels c/d = canonical `fig2cd_symmetric_forms.py` functions, 1:1) + `gen_S9_torus_construction.py`; data `06_engines/z3_hexagonal_P0/P0_PM_torus_trace_shape_classes_Ly2Ly3.csv`, `…_Ly5.csv` (short exact counts, sub-law branch) + `TSDHC_long_trace{,_arith5}.csv` (long chains)
Fig. 3e planar-disk markers (S6.3)	`03_data/uniform/disk_counts_exact.csv` (exact disk counts, binary R = 0–7 and arithmetic R = 0–4, with anchors and fit summary); engines in `04_scripts_L2/disk/`: `disk_count.c` (C, `__int128`, column-recursion DP, scatter/gather twin modes; `gcc -O3 -std=gnu11 -fopenmp`), `disk_lib.py` (reference implementation: torus/disk construction, DFS, frontier DP, column DP, torus column trace), `blocked_multi.py` (blocked-DFS cross-check); torus-side anchors against the vendored `metric_hc.py`; alignment document available from the authors (2026-10-15)
Fig. 4 (diffraction & anisotropy)	`02_figures/fig4_value_domain_halos.*`; script `fig4_value_domain_halos.py` (seven-panel v2, 2026-10-12; + modules `vd_suite3.py`, `vd_patch.py`, `vd_suite2.py` (helper), external `sampler2.py`/`metric_hc.py`, `P0_L1_halo_anisotropy.py`); data `03_data/uniform/` (`meanF2_*.npy`, `spots_*.csv` (per-spot f_coh/f_inc/f_coh_al/f_pk + bootstraps), `hscan_summary.csv`, `displacement_spectra.csv`, `wrms_curve_tor21.csv` (single-ensemble w_{rms}, panel d), `ens_*.npz`; raw per-configuration ensembles + sampler metadata in `03_data/uniform/raw/` (incl. `tor21_*.npz` and the h-scan `torus13_a123_h∓0.15/±0.30.npz` for the panel-e schematic)); server drivers `vd_torus_run.py` (ensemble generation), `vd_wrms.py` (patch-radius ladder)
Fig. S1 (model & closure)	`02_figures/S1_model_closure.*`; script `gen_SI_figs.py` (drawn schematic)
Fig. S2 (mechanisms)	`02_figures/S2_mechanisms.*`; script `gen_SI_figs.py`; kagome symbol from `06_engines/z4_kagome/KAG_zero_entropy_theorem_analytic_proof.md` (formula verified)
Fig. S3 (Λ saturation)	`02_figures/S3_lambda_saturation.*`; data `06_engines/z3_488/TSQ_Lambda_DP_exp.csv`
Fig. S4 (tile enumeration)	`02_figures/S4_tile_enumeration.*`; enumeration = fig2 `tile_orbits` (45 → 15 orbits, asserted)
Fig. S5 (resonant sectors)	`02_figures/S5_resonance_sectors.*`; data `06_engines/z3_hexagonal_P0/P0_PM_torus_trace_shape_classes_Ly2Ly3.csv`, `…_Ly5.csv` (resonant rows) + `TSDHC_long_trace{,_arith5}.csv` (long chains)
Fig. S6 (3.4.6.4 area law)	`02_figures/S6_area_law_3464.*`; script `figSI_area_law.py`; data `06_engines/z4_3464/RTH_torus_area_law_8pts.csv`
Fig. S7 (4.6.12: ternary series + spectral determination)	`02_figures/S7_4612_spectral.*`; generator `figS7_4612_spectral.py` (canonical copy; supersedes `06_engines/z3_4612/roundG_4612_lnOmega_plot.py` for this figure); data `06_engines/z3_4612/RTH4612_knuth_estimate_series.csv` (panel a), rate values from `06_engines/z3_4612/RTH4612_metric_entropy_TM_readjudication_2026-10-06.md` §1 rate table (incl. the 2026-10-09 W = 4 row) + logs `RTH4612_t4k1_trunk_2026-10-09.log` / `RTH4612_t4k5_rssabort_2026-10-09.log` (panel b; rates asserted as constants in the generator)

Item	Data / code
Fig. S8 (lower-bound certificates, SI S8.3)	`02_figures/S8_lower_bound_certificates.*`; composer `figS8_lower_bound_certs.py`; source renderings `02_figures/assets_certificates/` (`flip_certificate.png`, `cert_3464.png`, `cert_31212.png`, `cert_4612.png`), drawn by `flip_certificate_fig.py` and `a4_cert_figs_all.py` (real-metric, edge length = edge value)
Fig. S9 (PDF + S_L)	`02_figures/S9_pdf_slq.*`; script `vd_si_sigs.py`; data `03_data/uniform/pdf_decay_suite.csv`, `pdf_curves_suite.npz`, `slq_suite.csv`, `slq_profiles_suite.npz`
Fig. S10 (halo azimuth)	`02_figures/S10_halo_azimuth.*`; script `vd_si_sigs.py`; data `03_data/uniform/halo_orient_suite.csv`, `halo_orient_lock_suite.csv`
Fig. S11 (entropy split)	`02_figures/S11_entropy_split.*`; script `fig_si_entropy_split.py` (enumerator `si_data.py`); data `03_data/uniform/si_entropy_split.csv`
Table S1 (disambiguation register)	criteria sources in the disambiguation study (literature review, 2026-09-27; available from the authors); S10.2 data `03_data/uniform/slq_suite.csv`, `slq_profiles_suite.npz`
Table S2 (classification)	theorem documents + verification CSVs in `06_engines/z4_square/`, `06_engines/z6_triangular/`, `06_engines/z4_kagome/`, `06_engines/z3_488/`, `06_engines/penrose/`, `06_engines/z4_3464/`, `06_engines/z5_snubsquare/`, `06_engines/z5_snubhex/`, `06_engines/z5_33344/`, `06_engines/z3_4612/`
Table S4 (TM spectra + mixed model)	`06_engines/z3_hexagonal_P0/TSDHC_tm_rho.csv`, `TSDHC_tm_rho_nonarith.csv`, `TSDHC_binary_spectral.csv`, `TSDHC_nonarith_spectral.csv`; counters `step8_hc_tm.py`, `step9_hc_tm_nonarith.py`, `step10_hc_binary_spec.py`, `step11_hc_s_fits.py`, `step12_long_trace.py` (+`step12b_parallel.py`), `step14_hc_processing.py`; legacy exact-count CSVs `P0_PM_torus_trace_shape_classes_Ly2Ly3.csv`, `…_Ly5.csv` (sub-law branch data, S6.3)
S6.3 commensurate-sector counts	`06_engines/z3_hexagonal_P0/P0_PM_resonant_families_Ly9.csv` (+ anchor-gate log `P0_PM_resonant_families_Ly9_anchor_gate.log`); counters `P0_PM_hc_trace.py` + parallel driver `resonant_width9_par.py`
S6.3 exact-size rule grid	`03_data/uniform/exact_size_rule_grid.csv` (delivered as `grid_results_server_final.csv`, development tree `_temp/D8b_sidon/`, 2026-10-16; 2,232 rows, 2,088 distinct alphabet–size points, 30 alphabets k = 2–8, L_y = 2–6, $L_x \leq 150$; per-row alphabet flags and counts, the criterion prediction re-derivable from the gcd and 3-AP columns); criterion working documents available from the authors (analytical proof in preparation)
Table S5 (3.4.6.4)	`06_engines/z4_3464/RTH_torus_area_law_8pts.csv`; counter `rth_torus_trace_sparse.py`
Table S6 (4.6.12 $\mathbb{Z}$ counts)	`06_engines/z3_4612/RTH4612_three_rank_dimR_dimQs3_dimSplit.csv` (ranks), `06_engines/z3_4612/RTH4612_constraint_matrix_layer_results_2026-10-01.md` (pipeline), `06_engines/z3_4612/RTH4612_knuth_estimate_series.csv` ($s_{\mathbb{Z}}$ series; head records column semantics, estimator version, and the L = 1, 2 anchor-validation runs, with pre-fix rows marked superseded); counters `roundD_4612_numeric.py`, `roundE_bucket_count.py`, `roundE_4612_paramdfs.py`, `roundE_4612_knuth.py`; binary width-scaling determination (S8.2): `06_engines/z3_4612/RTH4612_binary_anchors.csv` (exact anchors, dual-engine), `06_engines/z3_4612/RTH4612_binary_strips.csv` (W = 2 frontier-DP series), `06_engines/z3_4612/RTH4612_binary_L4x3_summary.csv` (+ `RTH4612_binary_L4x3_blocks.csv`, depth-split 4×3), `06_engines/z3_4612/RTH4612_knuth_strips2.csv` (strip-Knuth series, extracted from `run_knuth_strips2.log`); counters `roundE_4612_binary_anchors.py`, `roundF_server_block$s_{bin}$ary.py`, `roundE_4612_frontier_dp.py`, `roundE_4612_strip_series.py`, `roundE_4612_knuth_strips2.py`; transfer-matrix spectral determination (S8.2): see the "Transfer-matrix determinations" row; exact settlement at L ≥ 5 open (S8.3 Q3)
Table S7 (value-domain ensemble suite)	`03_data/uniform/`: `meanF2_*.npy`, `spots_*.csv` (per-spot f_coh/f_inc/aniso + bootstraps), `ens_*.npz`, `displacement_spectra.csv`, `hscan_summary.csv`, `wrms_curve_tor21.csv`, `pdf_*`, `slq_*`, `halo_orient_*`; raw configurations in `03_data/uniform/raw/` (incl. `tor21_*.npz`, `torus13_a123_h∓0.15/±0.30.npz`, and the ladder extensions `torus13_a678.npz`, `torus13_a789.npz`); generators `vd_suite3.py`, `vd_ladder_ext.py`, `vd_wrms.py`, server drivers `vd_torus_run.py`, `vd_torus_run_v2.py` (dev), vendored `sampler2.py`, `metric_hc.py`; revision-round tables `03_data/uniform/ladder_ext.csv` (extended rungs) and `03_data/uniform/size_scaling_tor21.csv` (patch-size series)
Table R1 (counting engines)	counters `06_engines/z3_hexagonal_P0/pseudocrystal_lib.py`, `06_engines/z3_hexagonal_P0/P0_PM_hc_trace.py`/`P0_PM_hc_trace_Ly5.py`, `06_engines/z4_3464/rth_torus_trace_sparse.py`, `06_engines/z4_3464/hutch_rth.py`, `06_engines/z3_488/tsq_lambda/T_builder_v2.py`, `06_engines/z3_4612/roundE_bucket_count.py` + `06_engines/z3_4612/roundE_4612_paramdfs.py`; calibrations `06_engines/z3_hexagonal_P0/P0_A2b_Knuth_calibration_v2.csv`, `06_engines/z3_488/TSQ_Lambda_DP_exp.csv`, `06_engines/z3_4612/dev_calibrations/hutch_locW*_result.json` (dev)

Item	Data / code
Transfer-matrix determinations (S5.3, S6.2, S8.2)	3.12.12: `06_engines/z3_31212/step7_tm.py`, `step7_cross.py`, `step7_deep.py`, `step12_rho_solution.py`, `step13_rho_method.py` + `TSD31212_tm_*.csv`, `TSD31212_symmetric_fit.csv`; honeycomb: `06_engines/z3_hexagonal_P0/step8_hc_tm.py`, `step9_hc_tm_nonarith.py`, `step10_hc_binary_spec.py`, `step11_hc_s_fits.py`, `step12_long_trace.py` + `step12b_parallel.py`, `step14_hc_processing.py` + `TSDHC_tm_rho*.csv`, `TSDHC_*_spectral.csv`, `TSDHC_long_trace{,_arith5}.csv`, `TSDHC_processing.csv`; 4.6.12: `06_engines/z3_4612/rth4612_tm.py` (+ `roundE_4612_knuth_ternary_strips.py`, run logs) + `06_engines/z3_4612/RTH4612_metric_entropy_TM_readjudication_2026-10-06.md`, `RTH4612_binary_positive_entropy_adjudication_audit_2026-10-05.md`, `RTH4612_TM_redo_recommendations_2026-10-06.md`
S8.3 constructive certificates and strip construction	certificate and verification scripts in `04_scripts_L2/`: `flip_hc_ternary.py` (honeycomb flower certificate, kernel anchors, square-lattice negative control), `flip_3464_31212.py` (3.4.6.4 and 3.12.12 patches), `flip_488lam_4612.py` (4.6.12 patch; its 4.8.8 section is not used by the paper), `h2l_strip.py` (dual-algorithm strip counting, interval criterion, offset tables), `h2l_construct.py` (three site sets; subset, injectivity and end-to-end count tests); records `03_data/lower_bounds/*.json` (result readouts and mode tables)
Shared library	`06_engines/z3_hexagonal_P0/pseudocrystal_lib.py`